

Cyber Security Course Syllabus

Module 1: Introduction to Cyber Security

- What is Cyber Security?
 - Importance of Cyber Security
 - Types of Cyber Threats
 - Cyber Security career paths
 - Ethical vs unethical hacking
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Module 2: Networking Fundamentals

- Computer networks basics
 - OSI & TCP/IP models
 - IP addressing & DNS
 - Ports & protocols (HTTP, HTTPS, FTP, SMTP)
 - Firewalls & routers
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Module 3: Operating Systems & Security

- Windows & Linux basics

- User management & permissions
 - File system security
 - Process & memory management
 - OS hardening techniques
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Module 4: Cyber Threats & Attacks

- Malware (virus, worm, trojan, ransomware)
 - Phishing & social engineering
 - DoS & DDoS attacks
 - Man-in-the-Middle attacks
 - Zero-day vulnerabilities
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Module 5: Cryptography & Data Security

- Basics of cryptography
- Symmetric & asymmetric encryption
- Hashing & digital signatures
- SSL/TLS & HTTPS
- Data protection techniques

Module 6: Web Application Security

- Common web vulnerabilities
- OWASP Top 10
- SQL injection
- Cross-site scripting (XSS)
- Cross-site request forgery (CSRF)

Module 7: Network Security

- Network monitoring
- Intrusion Detection & Prevention Systems (IDS/IPS)
- VPNs & secure communication
- Wireless security (Wi-Fi attacks)
- Network access control

Module 8: Ethical Hacking & Penetration Testing

- Ethical hacking methodology
- Reconnaissance & scanning

- Vulnerability assessment
 - Exploitation basics
 - Post-exploitation concepts
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Module 9: Cloud & IoT Security

- Cloud computing basics
 - Cloud security challenges
 - Securing AWS/Azure/GCP
 - IoT threats & protection
 - Container & API security
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Module 10: Cyber Laws & Compliance

- Cyber laws & IT Act
 - GDPR & data privacy basics
 - Cyber ethics
 - Compliance standards (ISO 27001)
 - Risk management
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Module 11: Incident Response & Forensics

- Incident response lifecycle
 - Digital forensics basics
 - Evidence collection
 - Log analysis
 - Disaster recovery planning
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Module 12: Career Preparation & Projects

- Hands-on labs
- Real-world case studies
- Mini & capstone projects
- Resume & interview prep
- Certification guidance (CEH, Security+, CISSP)

Cyber Security Syllabus (7 Months)

FOUNDATION LEVEL (2 MONTHS)

Month 1: Cyber Security & Networking Basics

- Introduction to Cyber Security
- Cyber threats & attack types

- Basics of computer networks
- OSI & TCP/IP models
- IP addressing, DNS, DHCP
- Ports & protocols (HTTP, HTTPS, FTP, SMTP)
- Introduction to Firewalls & VPN
- Career paths in Cyber Security

Outcome: Understand how systems & networks work

Month 2: Operating Systems & Security Basics

- Windows security fundamentals
- Linux fundamentals for security
- User permissions & access control
- File system security
- System hardening basics
- Introduction to cryptography
- Password security & authentication
- Security best practices

Outcome: Strong system & OS security foundation

ADVANCED LEVEL (5 MONTHS)

Month 3: Cyber Threats, Attacks & Defense

- Malware analysis basics
- Phishing & social engineering attacks
- DoS & DDoS attacks
- Man-in-the-Middle attacks
- Vulnerability assessment tools
- Security monitoring basics
- Threat intelligence concepts

Outcome: Identify & respond to real-world threats

Month 4: Web & Network Security

- Web application architecture
- OWASP Top 10
- SQL Injection & XSS
- CSRF & authentication flaws
- Network security tools
- IDS / IPS
- Wireless security (Wi-Fi attacks)
- Network traffic analysis

Outcome: Secure websites & networks

Month 5: Ethical Hacking & Penetration Testing

- Ethical hacking methodology
- Reconnaissance & footprinting
- Scanning & enumeration
- Exploitation basics
- Password cracking techniques
- Social engineering simulations
- Post-exploitation concepts
- Reporting & documentation

Outcome: Hands-on ethical hacking skills

Month 6: Cloud, IoT & Advanced Security

- Cloud computing fundamentals
- Cloud security risks
- Securing AWS / Azure basics
- API security
- IoT security challenges
- Container & DevSecOps basics
- Zero Trust security model

Outcome: Secure modern & cloud systems

Month 7: Incident Response, Forensics & Career

- Incident response lifecycle
- Digital forensics basics
- Log analysis & SIEM basics
- Data privacy & cyber laws
- Risk management
- Capstone project
- Resume & interview preparation
- Certification guidance (CEH, Security+, CISSP)

Outcome: Job-ready cyber security professional